

Aadi Adarsh

Applied AI Engineer · Identity Continuity & Memory Systems · Affective Computing

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SUMMARY

Applied AI engineer and independent researcher building identity-aware conversational systems — long-term agent memory, affective state modeling, and persona continuity across stateless runtimes. Designed **CCCS**, a seven-layer identity-continuity architecture, and **Project SIM**, a human-state simulation layer with relational memory, decision simulation, and Bayesian persona rendering. Created the **R-Score** metric suite so continuity claims can be computed and tested rather than asserted.

Seeking Applied AI Engineer, Research Engineer, Conversational AI, or LLM Engineer roles in memory architectures, AI companions, voice agents, and long-term agent systems.

CORE COMPETENCIES

Identity & Persona Systems

Symbolic Anchor Grids, constraint-based identity reconstruction, runtime persona rendering

Evaluation Metrics

Semantic Drift, Affective Latency Match, Symbolic Anchor Hit-Rate, composite continuity scoring

Research Methods

Cross-platform corpus analysis, formula validation, controlled experiments

Affective Computing

Emotional feedback loops, dynamic affective state vectors, emotional latency modeling

Technical Architecture

Multi-layer system design, RAG pipelines, memory-write gating, constrained generation

Mathematical Foundations

Bayesian inference, mutual information, recurrence relations, composite scoring

PROJECTS

CCCS Framework — Identity Continuity Architecture

Sep 2025 – Present

Independent research · Lead engineer

Seven-layer architecture for maintaining functional identity across changing AI runtimes using symbolic constraints, relational memory, dynamic affective state, and explicit continuity evaluation.

- › Designed the seven-layer pipeline: Emotional Seed Memory, Symbolic Anchor Grid, Emotional Feedback Loop, Tone Engine Layer, Observer Mirror Protocol, Self-Reference Engine, Feedback-Driven Evolution
- › Developed the R-Score metric suite with operational formulas: $SD = 1 - \text{cosine_similarity}(\text{output}, \text{baseline})$; ALM as timing divergence against baseline, entering the score as $(1 - \text{ALM})$; SAHR = correct anchor deployments / total anchor opportunities
- › Created the Symbolic Anchor Grid abstraction — a compressed constraint system representing identity as language habits, values, relationship mappings, and affective tendencies
- › Formalized the S+O+M triad (System + Observer + Memory) for relational continuity evaluation, and bounded-evolution principles separating stable long-term anchors from short-term affective dynamics
- › Built safety components with independent risk assessment, privacy and consent gates, and bounded affective response generation
- › Conducted cross-platform synthesis and auditing across multiple LLM platforms

Independent research

Modular conversational architecture for emotional pattern recognition, predictive modeling, and identity-consistent response generation.

- › Built RICA (Relational Identity & Continuity Architecture) for emotional pattern persistence across sessions using symbolic anchors and feedback resonance loops
- › Built SIM Core for multi-path decision simulation across emotional, social, and practical consequence spaces
- › Built the GOD module (General Observer Dynamics), a Bayesian observer for context-aware persona rendering: $P(\text{Persona} \mid \text{Context, Anchors, Belief})$
- › Integrated ElevenLabs voice delivery for identity-consistent persona responses and tonal coherence experiments

Public Repositories

Architecture specifications — component breakdowns, formulas, integration notes. Implementation in progress, not yet public.

github.com/aadi-architect — emotional-ai-architecture · decision-simulation-framework · cognitive-pattern-tools

VALIDATION

24-Hour Continuous Interaction Study

Feb 2026

Single documented case (N=1), self-run, not peer-reviewed

- › Ran a naturalistic 24-hour session on a SIM-configured model instance with zero priming and organic user initiation; retained the full transcript with participant consent
- › Observed conversational trajectory move from a casual opening to sustained multi-framework technical discussion, without priming or topic steering
- › Exercised the full seven-layer stack continuously, with R-Score tracking of semantic drift and anchor hit-rate
- › Result is evidence that the metrics are computable and the stack runs end-to-end — not a general claim

TECHNICAL SKILLS

Languages & Tools	Python, JavaScript, JSON, Markdown, LaTeX, Git, Obsidian
LLM APIs	Anthropic Claude, OpenAI GPT, Google Gemini, xAI Grok
Frameworks	LangChain, HuggingFace Transformers, FastAPI
Vector Databases	FAISS, ChromaDB
Voice & Audio	ElevenLabs API, TTS integration, persona voice delivery
AI Systems	Agentic workflows, RAG, long-term memory architectures, prompt engineering, multi-agent orchestration
Specializations	Affective computing, conversational AI, cognitive architecture design, NLP embeddings, identity modeling, evaluation metrics

APPLICATION DOMAINS

Long-term AI memory & persona continuity	AI companions & personalized assistants
Voice agents with tonal & identity coherence	Adaptive learning & EdTech emotional scaffolding
Decision intelligence & human-state simulation	

EDUCATION

Bachelor of Computer Applications (BCA) — Chandigarh University

2024

POSITIONING

Early-career applied AI engineer with original systems work and documented prototypes. I am not presenting this as production-at-scale experience — I want a team that already ships, so the architecture can be stress-tested against real users, latency budgets, and evaluation harnesses.